Super El Niño winters in Finland and the Nordic countries

Temperature, precipitation, snow and frost in super El Niño winters compared with ordinary El Niño, neutral and La Niña winters, 1950 to 2025/26

Analysis report

September 4, 2026

Abstract

Question. Are Nordic winters warmer and wetter in super El Niño years (event peak Oceanic Niño Index, ONI, at or above 2.0 °C) than in ordinary El Niño years? **Data.** Daily observations from 13 Finnish Meteorological Institute stations (Helsinki Kaisaniemi from 1900, most from 1959) and 23 GHCN-Daily stations in Sweden, Norway, Denmark and Iceland, with ERA5 reanalysis (Open-Meteo) as a partial cross-check. Winters were classified from the NOAA ONI: 3 super El Niño winters (1982/83, 1997/98, 2015/16), 20 other El Niño winters, 29 neutral and 25 La Niña winters. **Method.** Seasonal (DJF, JFM) means and counts per station and winter; anomalies against 1991 to 2020; a linear trend removed per station, season and metric so that the late-record clustering of super events does not masquerade as an ENSO signal; group means with bootstrap confidence intervals and permutation tests. **Result.** In December to February the three super El Niño winters were milder than neutral winters in Finland, Sweden, Norway and Denmark once the warming trend is removed (Nordic mean 0.9 °C, 95 % interval -0.1 to 1.9; Finland 1.0 °C, -0.8 to 2.8), with more thaw days and slightly less snow, and Iceland was the exception with a small cooling. Against ordinary El Niño winters the temperature difference was smaller (Nordic 0.7 °C, -0.5 to 1.8). Finland was the only country with a clear precipitation surplus (26.3 mm over the season, 10.0 to 42.6). None of these differences reached $p < 0.05$ in a permutation test (Nordic temperature $p = 0.37$; Finnish precipitation $p = 0.11$), the January to March window showed no signal at all, and the ONI had no monotonic relation with any metric (Spearman $|\rho| \leq 0.15$). In absolute terms the three super winters were ordinary Finnish winters, not mild ones; the mildness appears only relative to the trend of their own era. **Caveat.** With three super El Niño winters, every interval is wide and no single-winter anecdote should be read as a rule. The interactive companion app lets the reader change the threshold, season, metric and region.

1 Background and hypothesis

El Niño alters the winter circulation over the North Atlantic and Europe, but the teleconnection is weak, seasonally shifting and not linear in event strength. Reviews of the European response find the most robust signal in late winter, when a warm tropical Pacific favours a weakened stratospheric polar vortex and a negative phase of the North Atlantic Oscillation, which in northern Europe means colder and drier rather than milder conditions [1, 2, 3, 4]. Early winter can show the opposite sign [1]. Model and reanalysis work on the strongest events suggests that the response saturates or even changes character for very strong El Niños, with a direct tropospheric route that projects less on the NAO [5, 6]. Seasonal forecast systems nevertheless draw real skill for European winters from the tropical Pacific [7].

The working hypothesis for this report was the everyday one: that super El Niño winters in Finland are milder and wetter than ordinary El Niño winters. The literature above gives a reason to expect the opposite in late winter, so the analysis was set up to let the data decide, with both DJF and JFM windows and with the same methods applied to all four ENSO groups.

2 Data

2.1 ENSO classification

The Oceanic Niño Index (three-month running mean of ERSSTv5 Niño 3.4 sea surface temperature anomalies) was downloaded from the NOAA Climate Prediction Center [8, 9]. Following the CPC definition, an event is at least five consecutive overlapping seasons with ONI at or above +0.5 °C (El Niño) or at or below −0.5 °C (La Niña). Each winter is labelled by the year of its January (winter 1997/98 is 1998) and takes the ONI of its DJF season. A winter inside an El Niño event whose peak ONI reaches 2.0 °C is a *super El Niño winter* only if it is the winter containing the peak; the second winter of a two-year event (2014/15) is classified on its own DJF value. Winters inside an event that never reaches 2.0 are *El Niño*; winters inside a La Niña event are *La Niña*; all others are *neutral*. Winters before 1950/51 carry no ONI label and were used only to estimate station trends and climatologies. The 2023/24 event peaked at 1.99 °C in the current ONI table and is therefore an ordinary El Niño winter at the 2.0 threshold; the companion app allows the threshold to be moved. At the time of writing (September 2026) a new El Niño is under way: the ONI reached 1.8 °C in June to August 2026 and NOAA gives a better than 90 % chance of a very strong event in the coming autumn and winter, so winter 2026/27 is the obvious first test of anything claimed here.

2.2 Station observations

Finnish daily observations (daily mean, maximum and minimum temperature, precipitation and snow depth) were fetched from the FMI open data WFS interface [10], one calendar year per request, for 13 stations chosen for record length and geographic spread (Table 4). The open interface serves most stations from 1959; Helsinki Kaisaniemi and Sodankylä Tähtelä extend to the early 1900s. At several airport stations precipitation and snow depth end around 2008 when observations moved to automatic systems; temperature continues. For Sweden, Norway, Denmark and Iceland daily data came from the NOAA Global Historical Climatology Network-Daily archive [11], which holds the national services' long series (Stockholm and Härnösand from the 1850s). Daily mean temperature was taken as reported (TAVG) or as the mean of TMAX and TMIN where TAVG is absent. Values with a quality flag were discarded. Denmark is thinly covered in GHCN-Daily after 2000; four stations were combined. Muonio Alamuonio (GHCN-Daily, 1959 to 2026, snow depth from 1982) was added as a single station because it is the nearest long record to Olos; it is shown at station level and in the app but is not part of the Finnish national series, which uses FMI stations only.

2.3 ERA5 reanalysis

The original plan used ERA5 through the Open-Meteo archive API for all non-Finnish points [12, 13]. Open-Meteo counts one call per two weeks of data per location, so an 86-year daily series costs about 2 260 calls against a free quota of 10 000 per day. ERA5 series were therefore fetched only for a few points as a cross-check, and the reported Nordic results rest on station observations.

2.4 Seasonal metrics

For each station, winter and window (DJF: December to February; JFM: January to March; NDJFM as a five-month window in the app) the following were computed when at least 90 % of the days were present: mean temperature; mean daily maximum and minimum; precipitation sum;

wet days ($\geq$1 mm); mean and maximum snow depth; snow-cover days ($\geq$1 cm); frost days (Tmin $<$ 0 °C); ice days (Tmax $<$ 0 °C); thaw days (Tmax $>$ 0 °C); and rain-on-snow days (liquid precipitation $\geq$1 mm on a day with snow cover, using rain where reported and otherwise precipitation on days with mean temperature above 0.5 °C). Counts were scaled to the full window length when a few days were missing.

3 Methods

Three versions of every metric are analysed: the raw value; the anomaly relative to the 1991 to 2020 mean of the same station, window and metric; and a *detrended anomaly*, the residual from an ordinary least squares line fitted through all available winters of that station, window and metric. The detrended version is the primary one. The three super El Niño winters fall in 1983, 1998 and 2016, late in a record with a warming trend of roughly 0.2 to 0.5 °C per decade in Nordic winters, so raw composites would credit El Niño with warming that belongs to the trend. National series are the mean of station anomalies per winter (FMI for Finland, GHCN-Daily elsewhere; at least two stations required), and a Nordic series is the mean of the five national series.

For each unit (station or country), window, metric and version, group means with 95 % confidence intervals are reported: a percentile bootstrap (4 000 resamples) for groups of six or more winters and a t-interval for smaller groups, because a bootstrap of three values has only ten distinct resamples and understates the uncertainty. Pairwise differences of means (super minus El Niño, super minus neutral, El Niño minus neutral, La Niña minus neutral) carry bootstrap intervals, or Welch t-intervals when either group is small, together with a two-sided permutation test of the difference in means (5 000 permutations), Welch's t and Mann-Whitney U as sensitivity checks, and Cohen's d. Spearman's rank correlation between the DJF ONI and the metric across all labelled winters is given as a continuous alternative to the four groups. The permutation test is the primary test: with three super winters it asks how often three winters drawn at random from the record differ from the rest by as much as the observed three, and it does not depend on the spread within the super group, which the t-based intervals do. No correction for multiple comparisons was applied; the number of tests is large and the reader should treat isolated significant cells as expected noise.

All code is in `src/` (Python 3.13: pandas, NumPy, SciPy, Matplotlib). The pipeline runs end to end with `python3.13 src/run_all.py`; the offline app is `app/index.html`.

4 Results

4.1 ENSO groups

The ONI record (Figure 1) yields 3 super El Niño winters (1982/83, 1997/98, 2015/16), 20 other El Niño winters, 29 neutral winters and 25 La Niña winters between 1950/51 and 2025/26. Peak ONI values were 2.14 (1983), 2.37 (1998) and 2.59 °C (2016); the next strongest events were 2023/24 (1.99), 1972/73 (1.84) and 1965/66 (1.82).

4.2 December to February

Temperature. Detrended DJF mean temperature in super El Niño winters exceeded neutral winters in every mainland Nordic country (Figure 2, Table 2): Finland 1.0 °C (-0.8 to 2.8, permutation $p = 0.52$), Sweden 1.6 °C (0.1 to 3.1, $p = 0.28$), Norway 1.2 °C (0.1 to 2.2, $p = 0.28$), Denmark 1.4 °C (0.6 to 2.1, $p = 0.16$) and the Nordic mean 0.9 °C (-0.1 to 1.9, $p = 0.37$). Iceland went the other way (-0.6 °C, -2.2 to 1.0). Ordinary El Niño winters were only marginally milder than neutral ones (Nordic 0.3 °C, -0.8 to 1.3, $p = 0.61$), so most of the super-versus-neutral difference is also present as a super-versus-El Niño difference (Nordic 0.7 °C, -0.5 to 1.8, $p = 0.60$; Finland 0.8 °C, -1.3 to 2.8, $p = 0.71$). La Niña winters did not differ from neutral winters in any country.

The permutation p values are the ones to read. Welch's t gives p between 0.04 and 0.07 for Sweden, Norway and the Nordic mean, but it estimates the spread of the super group from three winters that happen to resemble each other; the permutation test asks how often three winters drawn at random from the record produce a mean this far from the rest, and the answer is one time in three to one time in seven. The intervals quoted above for the super group are Welch t-intervals and share that fragility.

Precipitation. Finland was the only country with a coherent precipitation surplus in super El Niño winters: 26.3 mm over DJF relative to neutral winters (10.0 to 42.6, $p = 0.11$) and 26.3 mm relative to ordinary El Niño winters (8.8 to 43.8, $p = 0.13$), about a quarter of the seasonal total, with the surplus present in all three winters (detrended anomalies of $+16$, $+23$ and $+34$ mm in 1983, 1998 and 2016). Norway showed a surplus of similar size with a far wider interval (36.4 mm, -53.8 to 126.5), Sweden and Denmark showed nothing (-1.5 and 2.0 mm), and Iceland was drier on two winters of data. The Nordic mean was 7.7 mm (-14.1 to 29.5). Ordinary El Niño winters did not differ from neutral winters in precipitation anywhere (Figure 4).

Snow, frost and thaw. Mean snow depth was lower in super winters (Nordic -3.8 cm, -6.7 to -1.0, $p = 0.38$; Finland -2.4 cm, -5.5 to 0.6; Figure 5). Thaw days were more frequent (Nordic 6.6 days, 1.3 to 11.8, $p = 0.26$; Finland 7.6 days, 1.9 to 13.2, $p = 0.26$; Figure 6) and frost days slightly fewer (Finland -2.5 days, -4.8 to -0.1). These are the same signal as the temperature difference seen through different thresholds, not independent evidence.

Spatial pattern. At station level (Figures 10 to 13) the warm anomaly is largest in southern Sweden, Denmark and southern Norway and fades northward; Icelandic stations are cool. The Finnish precipitation surplus is present at every FMI station with a complete record. Only a handful of the several hundred station-by-metric cells reach $p < 0.05$, consistent with chance in an uncorrected table.

Absolute values. The three super winters were not mild in absolute terms. National DJF means in Finland were -6.5, -7.6 and -6.2 °C in 1982/83, 1997/98 and 2015/16, that is anomalies of $+0.3$, -0.6 and $+0.6$ °C against 1991 to 2020 (Figure 7, Supplementary Figure S1 in `graphs_export/`). They rank as mild only relative to the trend line of their own decade, and the detrended framing is what makes them look warm.

4.3 January to March

In JFM the temperature signal disappears (Table 3): Finland -0.3 °C, Sweden $+0.6$, Norway $+0.4$, Denmark $+1.0$, Iceland -0.8, Nordic mean $+0.2$ °C, all with intervals spanning zero and permutation $p \geq 0.3$ (Figure 3). The Finnish precipitation surplus shrinks to $+18$ mm ($p = 0.26$). Ordinary El Niño winters are, if anything, slightly cooler than neutral in JFM in Finland (-0.35 °C), in the direction the late-winter stratospheric pathway predicts, but far from distinguishable from noise.

4.4 Continuous ONI

Across all labelled winters the DJF ONI had no monotonic relation with any national metric: Spearman's ρ for DJF temperature was between -0.02 and 0.09 and for precipitation between -0.15 and 0.07 (Figure 9). Whatever the three super winters share, it is not the tail of a linear dose-response.

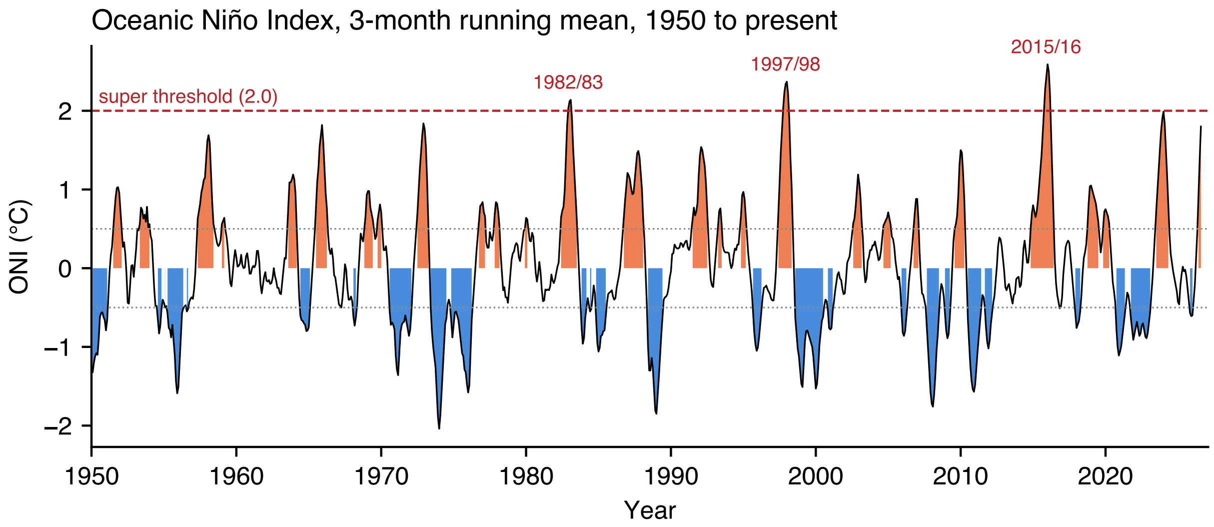

Figure 1: Oceanic Niño Index 1950 to present. Warm (orange) and cold (blue) shading marks values beyond $\pm0.5\,°C$. The three winters whose events peaked at or above 2.0 °C are labelled.

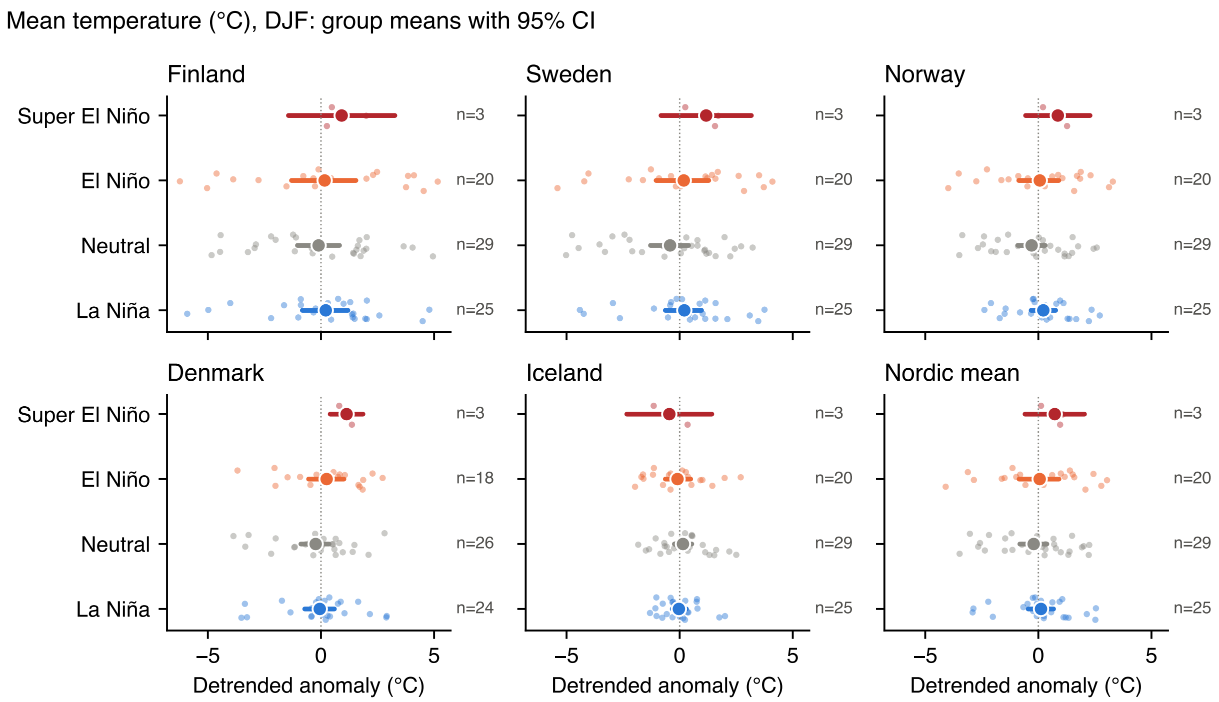

Figure 2: DJF mean temperature, detrended anomaly, by ENSO group and region. Dots are group means, bars 95 % confidence intervals (percentile bootstrap; t-interval for the three super winters), small points individual winters.

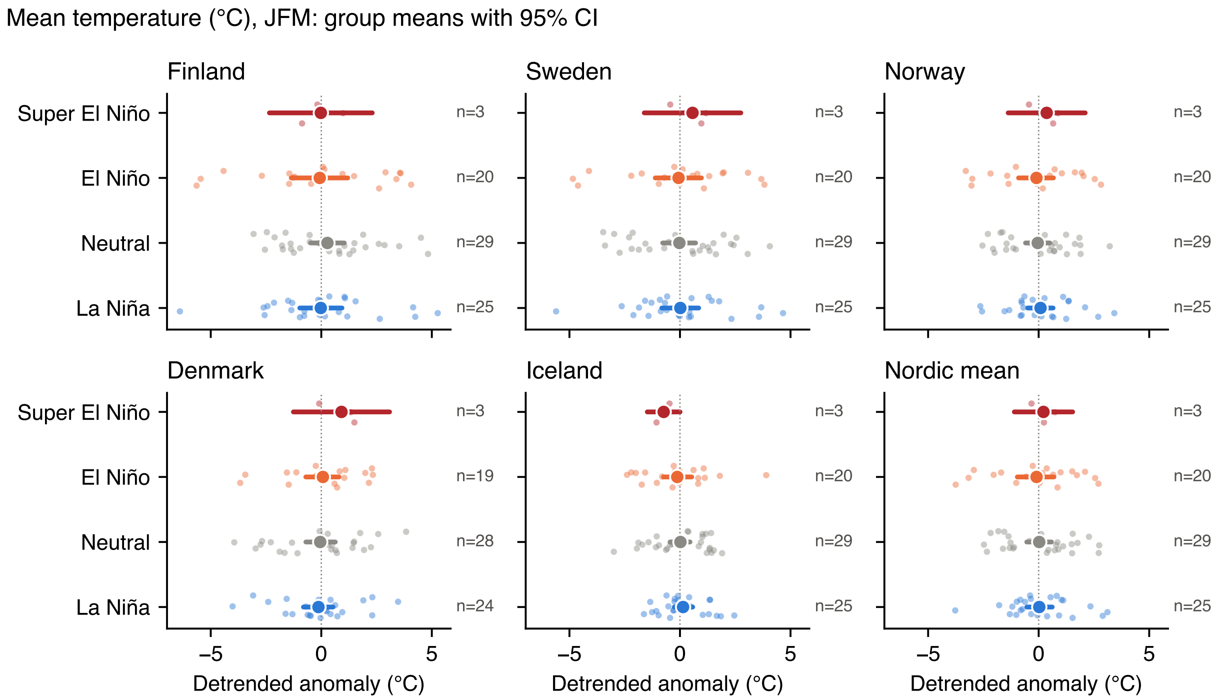

Figure 3: As Figure 2 for January to March.

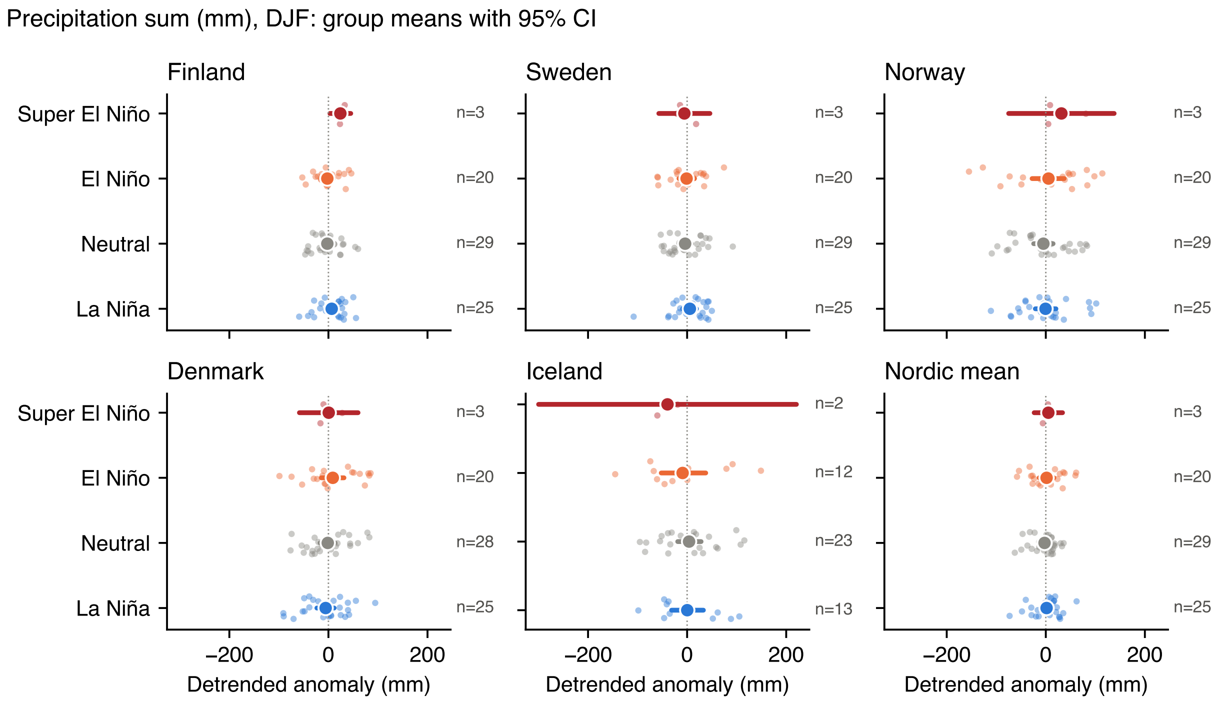

Figure 4: DJF precipitation sum, detrended anomaly, by ENSO group and region.

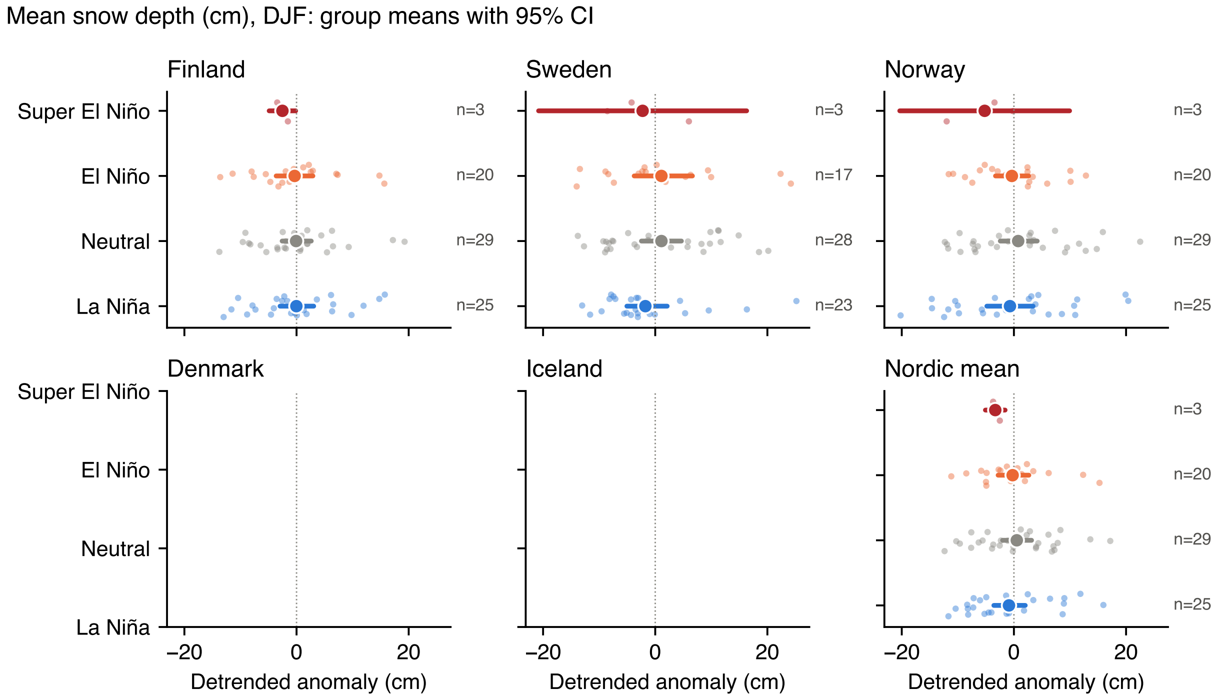

Figure 5: DJF mean snow depth, detrended anomaly, by ENSO group and region. Snow depth is complete for Finland (FMI) and for most Norwegian and Swedish GHCN stations; Denmark and Iceland have few snow-depth observations and their panels are sparse or empty.

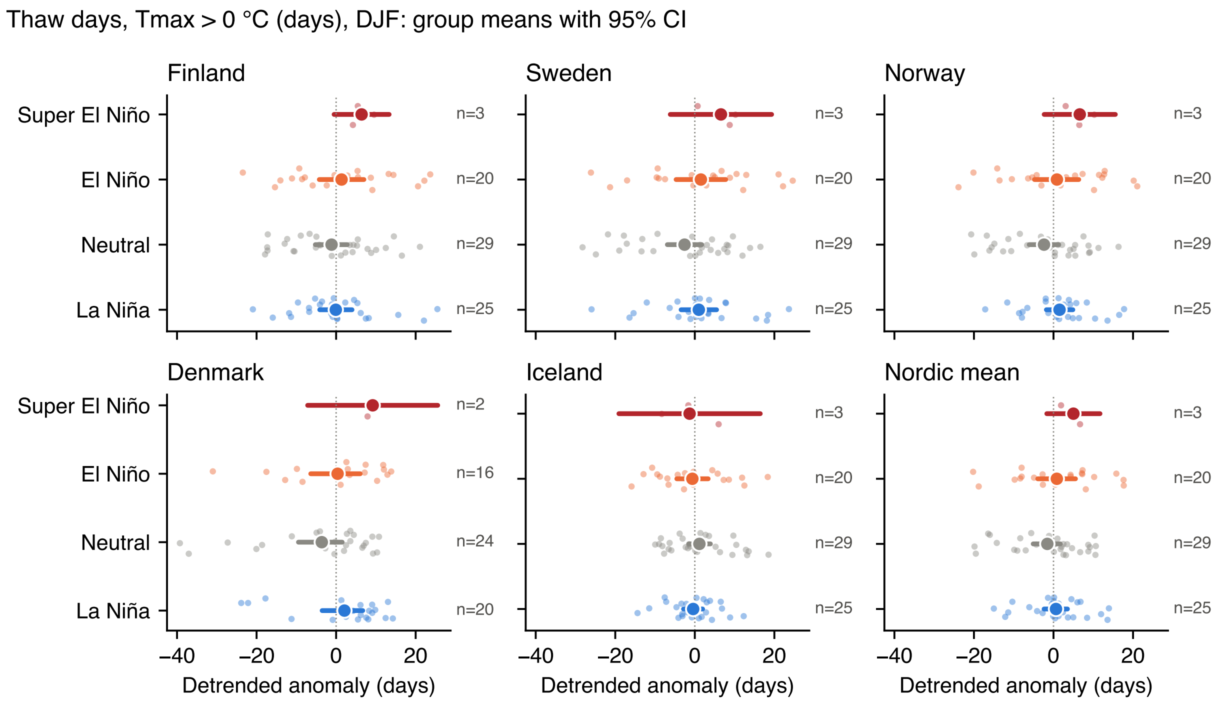

Figure 6: DJF thaw days (days with maximum temperature above 0 °C), detrended anomaly, by ENSO group and region.

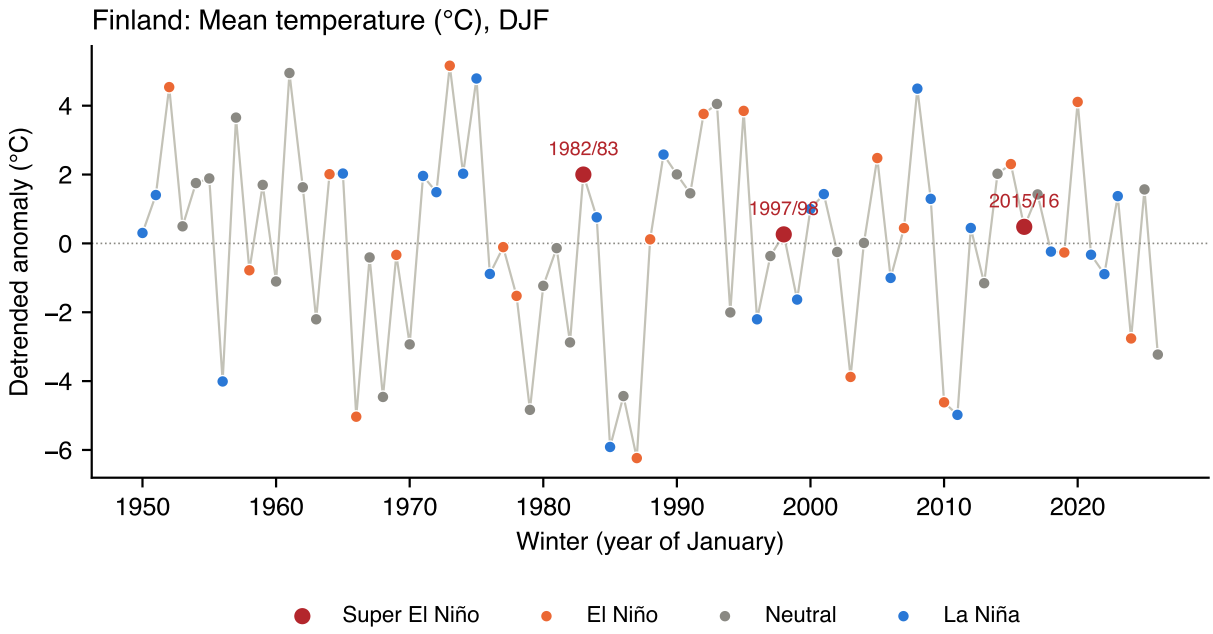

Figure 7: Finland, national mean of FMI stations: DJF mean-temperature detrended anomaly per winter, coloured by ENSO group.

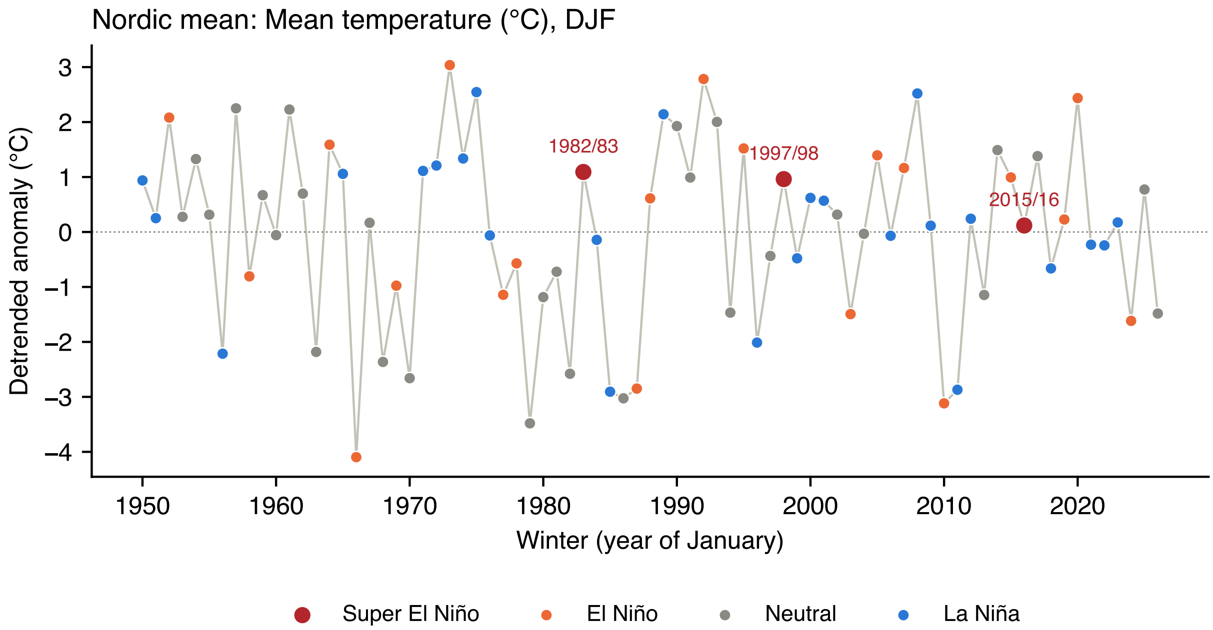

Figure 8: Nordic mean (five national series): DJF mean-temperature detrended anomaly per winter.

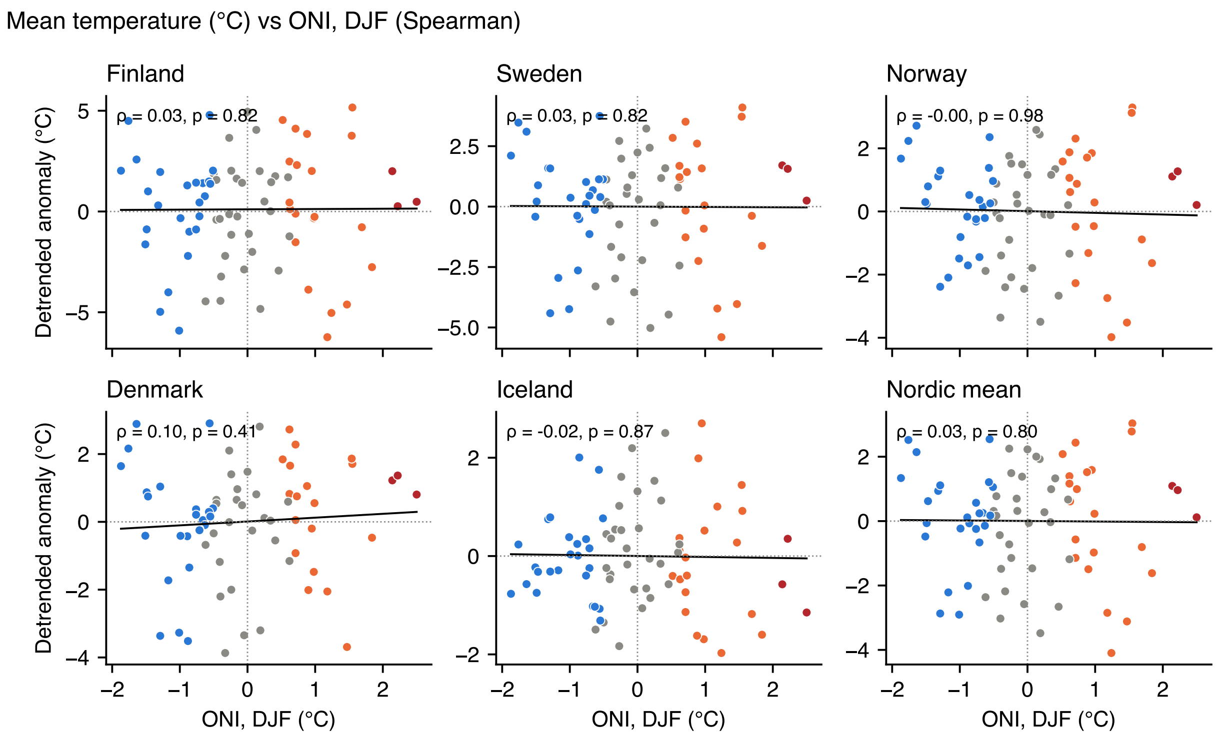

Figure 9: DJF mean-temperature detrended anomaly against the DJF ONI, all labelled winters, with an ordinary least squares line and Spearman's ρ.

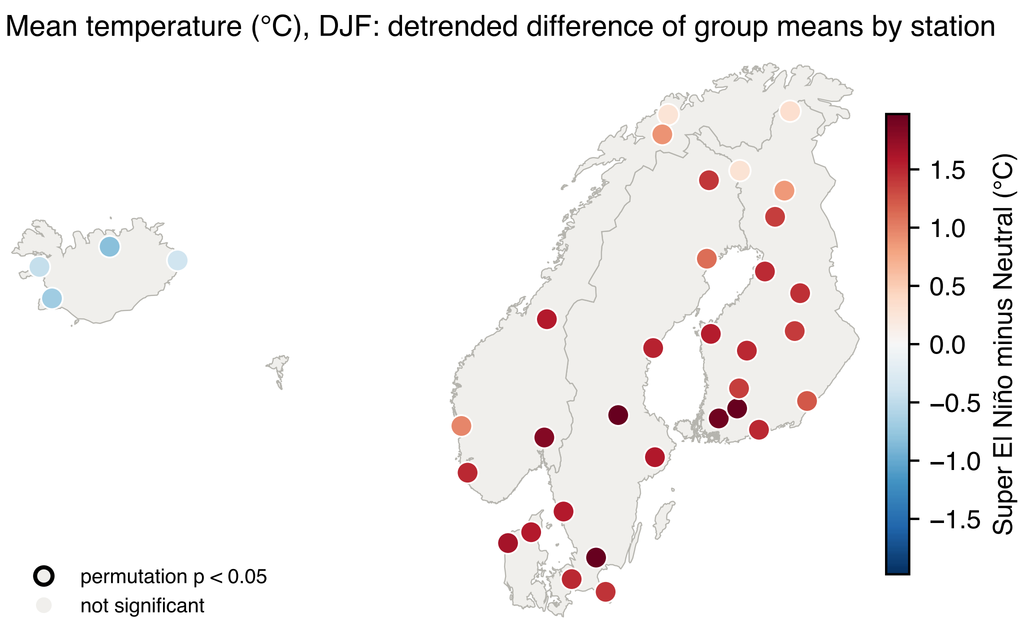

Figure 10: Station-level difference in DJF mean temperature (detrended anomaly), super El Niño minus neutral winters. Black outline: permutation $p < 0.05$.

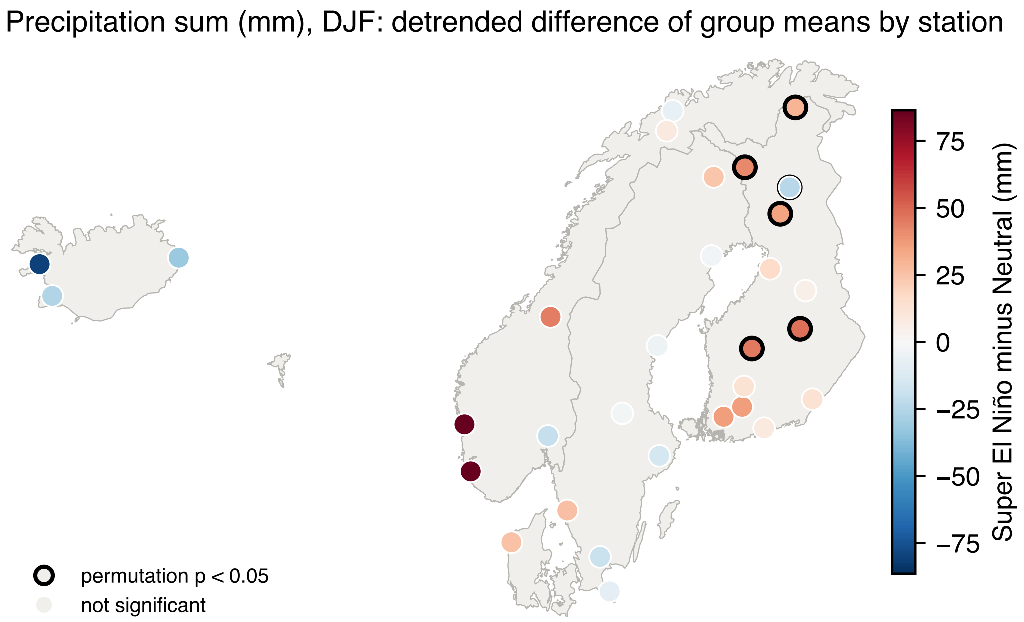

Figure 11: As Figure 10 for DJF precipitation sum.

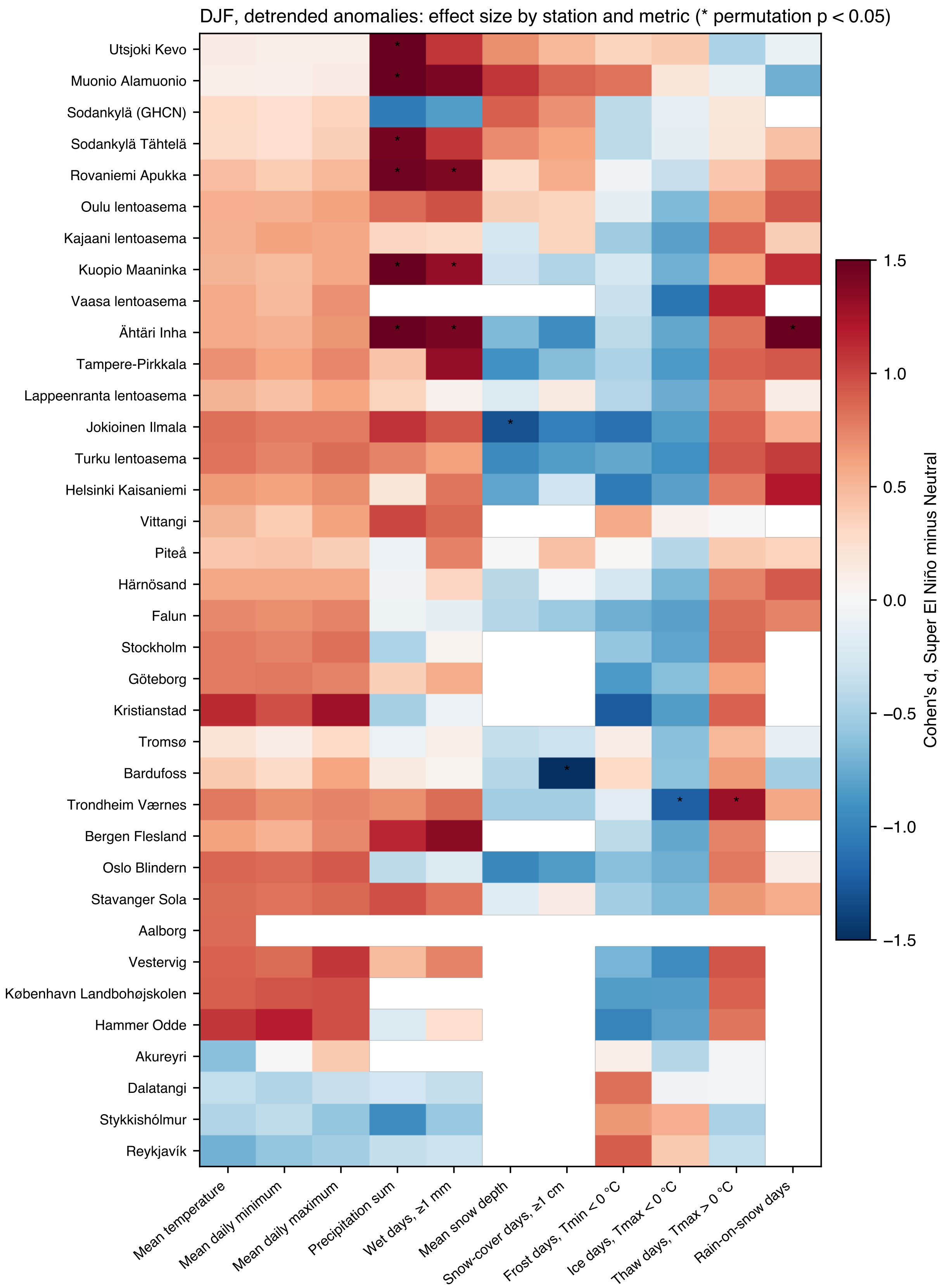

Figure 12: Effect size (Cohen's d) of super El Niño minus neutral winters, DJF detrended anomalies, by station and metric. Asterisks mark permutation $p < 0.05$; with $n = 3$ these are uncorrected and expected to occur by chance in a table of this size.

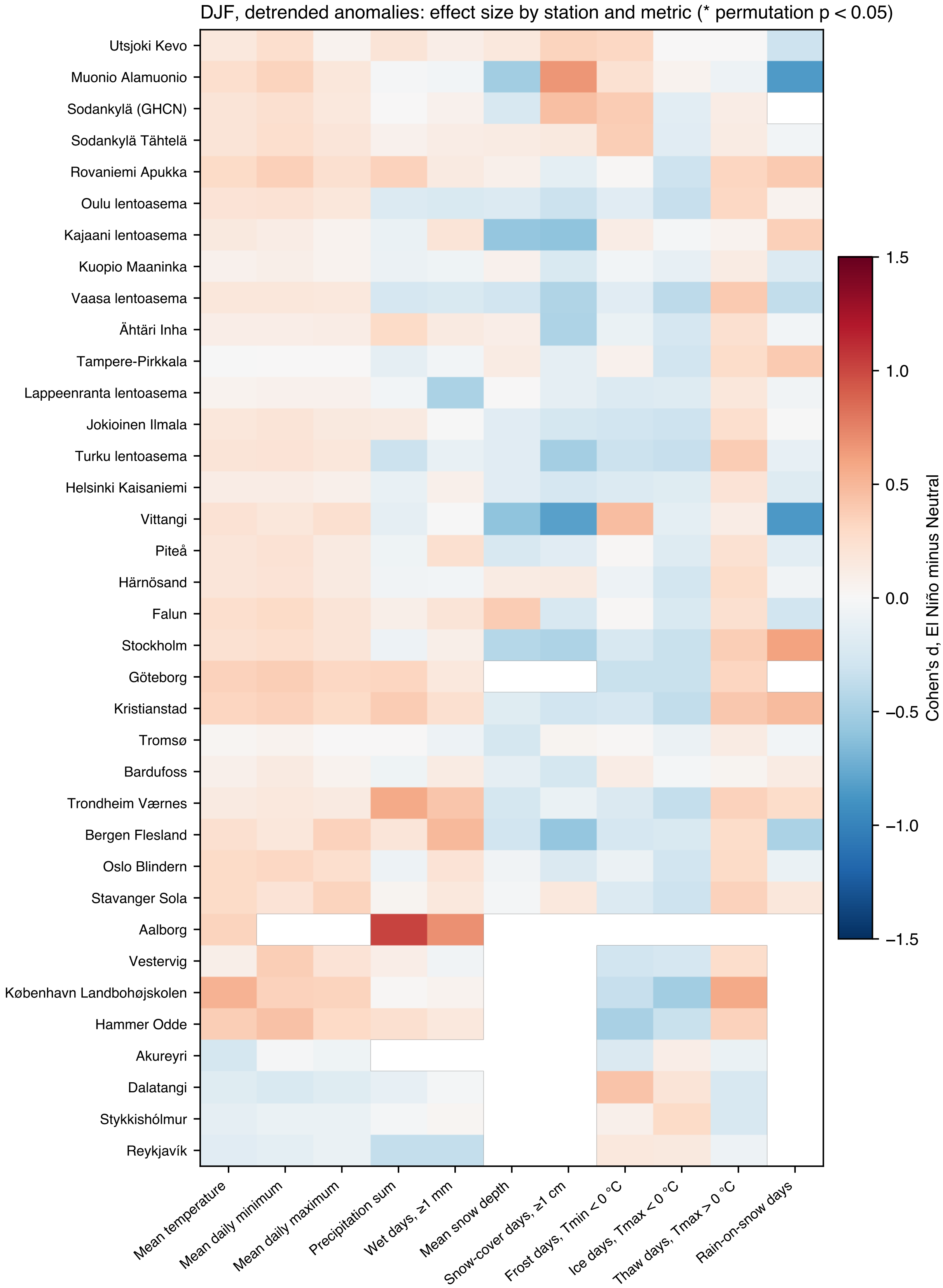

Figure 13: As Figure 12 for ordinary El Niño minus neutral winters.

Table 1: DJF detrended anomalies: group means with 95 % confidence intervals (percentile bootstrap; t-interval when $n < 6$), national series. Units: temperature °C, precipitation mm, snow depth cm, days.

Metric	Region	Super El Niño	El Niño	Neutral	La Niña
Mean temperature (°C)	Finland	0.9 (-1.4 to 3.3)	0.2 (-1.3 to 1.5)	-0.1 (-1.0 to 0.8)	0.2 (-0.8 to 1.2)
	Sweden	1.2 (-0.8 to 3.2)	0.2 (-1.0 to 1.3)	-0.4 (-1.3 to 0.4)	0.2 (-0.6 to 1.0)
	Norway	0.9 (-0.6 to 2.3)	0.1 (-0.9 to 0.9)	-0.3 (-0.9 to 0.3)	0.2 (-0.3 to 0.8)
	Denmark	1.1 (0.4 to 1.9)	0.3 (-0.5 to 1.0)	-0.2 (-0.9 to 0.4)	-0.1 (-0.7 to 0.6)
	Iceland	-0.5 (-2.3 to 1.4)	-0.1 (-0.6 to 0.5)	0.1 (-0.2 to 0.5)	-0.0 (-0.3 to 0.3)
	Nordic mean	0.7 (-0.6 to 2.0)	0.1 (-0.9 to 0.9)	-0.2 (-0.8 to 0.4)	0.1 (-0.5 to 0.7)
Precipitation sum (mm)	Finland	24.1 (3.1 to 45.1)	-2.2 (-13.8 to 9.8)	-2.2 (-12.4 to 7.8)	6.6 (-4.8 to 17.9)
	Sweden	-5.5 (-57.1 to 46.0)	-1.1 (-16.4 to 14.8)	-4.0 (-16.2 to 8.6)	5.5 (-9.0 to 18.8)
	Norway	31.5 (-75.0 to 138.0)	5.4 (-27.7 to 37.1)	-4.9 (-24.5 to 14.7)	-0.6 (-21.1 to 20.5)
	Denmark	0.6 (-58.6 to 59.8)	8.8 (-14.5 to 31.4)	-1.4 (-16.9 to 13.5)	-5.6 (-23.3 to 11.6)
	Iceland	-39.6 (-299.8 to 220.6)	-9.1 (-52.2 to 37.6)	3.9 (-19.0 to 28.2)	0.3 (-31.3 to 32.9)
	Nordic mean	5.1 (-23.5 to 33.8)	1.6 (-13.4 to 16.4)	-2.5 (-12.3 to 6.5)	1.6 (-10.2 to 12.8)
Mean snow depth (cm)	Finland	-2.5 (-4.9 to -0.2)	-0.3 (-3.6 to 2.9)	-0.1 (-2.6 to 2.6)	-0.0 (-2.9 to 3.1)
	Sweden	-2.2 (-20.8 to 16.3)	1.1 (-3.8 to 6.6)	1.1 (-2.5 to 4.7)	-1.8 (-5.1 to 2.1)
	Norway	-5.2 (-20.4 to 9.9)	-0.4 (-3.3 to 2.6)	0.8 (-2.4 to 4.2)	-0.7 (-4.8 to 3.4)
	Nordic mean	-3.3 (-5.1 to -1.6)	-0.2 (-2.8 to 2.7)	0.5 (-2.0 to 3.1)	-0.9 (-3.6 to 2.0)
Thaw days, Tmax $>$ 0 °C (days)	Finland	6.4 (-0.5 to 13.3)	1.4 (-4.2 to 7.0)	-1.1 (-5.2 to 2.8)	-0.1 (-4.1 to 4.0)
	Sweden	6.6 (-6.2 to 19.3)	1.5 (-4.7 to 7.8)	-2.6 (-6.9 to 1.8)	1.0 (-3.4 to 5.5)
	Norway	6.6 (-2.4 to 15.6)	0.9 (-4.8 to 6.4)	-2.4 (-6.1 to 1.3)	1.5 (-1.8 to 4.9)
	Denmark	9.2 (-7.2 to 25.5)	0.4 (-6.4 to 6.1)	-3.6 (-9.5 to 1.5)	2.1 (-3.5 to 6.7)
	Iceland	-1.3 (-19.1 to 16.5)	-0.6 (-4.5 to 3.4)	1.1 (-1.5 to 3.9)	-0.4 (-2.8 to 2.0)
	Nordic mean	5.0 (-1.7 to 11.7)	0.8 (-4.0 to 5.5)	-1.6 (-5.1 to 1.7)	0.6 (-2.4 to 3.5)
Frost days, Tmin $<$ 0 °C (days)	Finland	-2.4 (-5.3 to 0.4)	-0.5 (-3.0 to 1.8)	0.0 (-1.8 to 1.6)	0.1 (-1.7 to 1.8)
	Sweden	-4.0 (-17.8 to 9.8)	-0.6 (-4.3 to 2.6)	0.8 (-1.8 to 3.4)	-0.1 (-2.9 to 2.8)
	Norway	-2.3 (-15.9 to 11.3)	-0.7 (-4.5 to 3.1)	0.9 (-2.3 to 3.9)	-0.2 (-2.9 to 2.4)
	Denmark	-10.0 (-27.7 to 7.6)	-1.8 (-9.1 to 6.6)	3.8 (-2.3 to 10.2)	-1.6 (-9.1 to 6.2)
	Iceland	5.2 (-4.3 to 14.6)	0.3 (-5.2 to 5.2)	-1.9 (-5.2 to 1.2)	1.3 (-2.0 to 4.5)
	Nordic mean	-2.7 (-12.9 to 7.4)	-0.6 (-3.7 to 2.6)	0.7 (-2.0 to 3.2)	-0.0 (-2.4 to 2.4)

Table 2: DJF detrended anomalies: differences of group means with 95 % confidence intervals (bootstrap; Welch t-interval when a group has $n < 6$) and permutation p (*: $p < 0.05$, uncorrected).

Metric	Region	Super − El Niño	Super − Neutral	El Niño − Neutral
Mean temperature (°C)	Finland	0.8 (-1.3 to 2.8), p=0.71	1.0 (-0.8 to 2.8), p=0.52	0.3 (-1.4 to 1.9), p=0.76
	Sweden	1.0 (-0.7 to 2.7), p=0.55	1.6 (0.1 to 3.1), p=0.28	0.6 (-0.9 to 2.0), p=0.42
	Norway	0.8 (-0.5 to 2.1), p=0.54	1.2 (0.1 to 2.2), p=0.28	0.4 (-0.7 to 1.4), p=0.50
	Denmark	0.9 (-0.0 to 1.8), p=0.42	1.4 (0.6 to 2.1), p=0.16	0.5 (-0.5 to 1.5), p=0.35
	Iceland	-0.4 (-1.8 to 1.1), p=0.65	-0.6 (-2.2 to 1.0), p=0.35	-0.2 (-0.9 to 0.4), p=0.47
	Nordic mean	0.7 (-0.5 to 1.8), p=0.60	0.9 (-0.1 to 1.9), p=0.37	0.3 (-0.8 to 1.3), p=0.61
Precipitation sum (mm)	Finland	26.3 (8.8 to 43.8), p=0.13	26.3 (10.0 to 42.6), p=0.11	0.0 (-15.3 to 15.6), p=1.00
	Sweden	-4.5 (-44.1 to 35.2), p=0.84	-1.5 (-42.6 to 39.6), p=0.94	2.9 (-16.5 to 22.7), p=0.78
	Norway	26.1 (-55.6 to 107.7), p=0.57	36.4 (-53.8 to 126.5), p=0.28	10.3 (-28.9 to 46.3), p=0.58
	Denmark	-8.2 (-53.2 to 36.8), p=0.79	2.0 (-44.5 to 48.5), p=0.93	10.2 (-17.4 to 38.1), p=0.46
	Iceland	-30.5 (-112.5 to 51.6), p=0.61	-43.5 (-153.1 to 66.0), p=0.31	-13.1 (-60.8 to 37.4), p=0.59
	Nordic mean	3.6 (-19.5 to 26.7), p=0.87	7.7 (-14.1 to 29.5), p=0.64	4.1 (-13.0 to 22.2), p=0.64
Mean snow depth (cm)	Finland	-2.2 (-5.9 to 1.5), p=0.61	-2.4 (-5.5 to 0.6), p=0.55	-0.3 (-4.6 to 4.0), p=0.90
	Sweden	-3.4 (-17.8 to 11.1), p=0.63	-3.4 (-18.8 to 12.1), p=0.59	-0.0 (-6.3 to 6.2), p=1.00
	Norway	-4.8 (-17.5 to 7.8), p=0.26	-6.0 (-18.3 to 6.3), p=0.31	-1.1 (-5.6 to 3.4), p=0.65
	Nordic mean	-3.1 (-6.2 to -0.0), p=0.41	-3.8 (-6.7 to -1.0), p=0.38	-0.7 (-4.4 to 3.1), p=0.70
Thaw days, Tmax $>$ 0 °C (days)	Finland	5.0 (-2.1 to 12.2), p=0.54	7.6 (1.9 to 13.2), p=0.26	2.5 (-4.3 to 9.6), p=0.47
	Sweden	5.0 (-4.9 to 15.0), p=0.56	9.2 (-0.5 to 18.8), p=0.23	4.1 (-3.3 to 11.8), p=0.28
	Norway	5.7 (-2.0 to 13.5), p=0.48	9.0 (2.2 to 15.8), p=0.15	3.2 (-3.6 to 9.7), p=0.33
	Denmark	8.8 (1.4 to 16.2), p=0.35	12.8 (6.0 to 19.6), p=0.20	4.0 (-4.5 to 12.2), p=0.38
	Iceland	-0.7 (-15.1 to 13.7), p=0.90	-2.5 (-18.0 to 13.1), p=0.63	-1.8 (-6.7 to 3.1), p=0.47
	Nordic mean	4.2 (-2.1 to 10.4), p=0.54	6.6 (1.3 to 11.8), p=0.26	2.4 (-3.3 to 8.3), p=0.41
Frost days, Tmin $<$ 0 °C (days)	Finland	-1.9 (-4.9 to 1.1), p=0.57	-2.5 (-4.8 to -0.1), p=0.36	-0.6 (-3.6 to 2.3), p=0.70
	Sweden	-3.4 (-14.3 to 7.5), p=0.50	-4.8 (-16.5 to 6.8), p=0.28	-1.4 (-5.9 to 2.8), p=0.52
	Norway	-1.6 (-12.1 to 8.9), p=0.77	-3.2 (-14.2 to 7.8), p=0.54	-1.6 (-6.3 to 3.4), p=0.53
	Denmark	-8.2 (-21.9 to 5.4), p=0.46	-13.8 (-27.2 to -0.5), p=0.19	-5.6 (-15.2 to 4.4), p=0.29
	Iceland	4.9 (-2.8 to 12.6), p=0.50	7.1 (-0.0 to 14.3), p=0.18	2.2 (-3.9 to 7.7), p=0.45
	Nordic mean	-2.1 (-9.9 to 5.6), p=0.65	-3.4 (-11.3 to 4.6), p=0.46	-1.2 (-5.4 to 3.0), p=0.56

Table 3: As Table 2 for January to March.

Metric	Region	Super − El Niño	Super − Neutral	El Niño − Neutral
Mean temperature (°C)	Finland	0.1 (-1.9 to 2.0), p=0.98	-0.3 (-2.1 to 1.5), p=0.82	-0.3 (-1.8 to 1.1), p=0.64
	Sweden	0.6 (-1.1 to 2.4), p=0.69	0.6 (-1.1 to 2.2), p=0.64	-0.0 (-1.3 to 1.2), p=0.95
	Norway	0.5 (-0.9 to 1.8), p=0.68	0.4 (-0.9 to 1.7), p=0.66	-0.1 (-1.0 to 0.9), p=0.91
	Denmark	0.8 (-0.8 to 2.5), p=0.44	1.0 (-0.7 to 2.6), p=0.39	0.1 (-0.9 to 1.1), p=0.82
	Iceland	-0.6 (-1.4 to 0.2), p=0.49	-0.8 (-1.4 to -0.1), p=0.33	-0.1 (-0.9 to 0.7), p=0.73
	Nordic mean	0.3 (-0.8 to 1.4), p=0.79	0.2 (-0.8 to 1.2), p=0.84	-0.1 (-1.1 to 0.8), p=0.80
Precipitation sum (mm)	Finland	13.8 (-14.3 to 41.8), p=0.33	17.9 (-10.0 to 45.9), p=0.26	4.2 (-9.8 to 17.5), p=0.57
	Sweden	-5.9 (-45.2 to 33.5), p=0.78	3.7 (-39.1 to 46.4), p=0.82	9.5 (-8.0 to 27.6), p=0.30
	Norway	4.9 (-155.0 to 164.8), p=0.88	8.0 (-154.4 to 170.4), p=0.82	3.1 (-26.0 to 33.1), p=0.84
	Denmark	-10.8 (-89.3 to 67.7), p=0.72	6.0 (-81.3 to 93.4), p=0.79	16.8 (-8.9 to 41.4), p=0.17
	Iceland	-15.5 (-579.3 to 548.2), p=0.77	-22.9 (-641.4 to 595.7), p=0.65	-7.3 (-46.8 to 31.8), p=0.73
	Nordic mean	-4.4 (-69.5 to 60.6), p=0.80	3.3 (-63.3 to 69.9), p=0.85	7.7 (-6.9 to 23.2), p=0.34
Mean snow depth (cm)	Finland	-2.6 (-6.8 to 1.6), p=0.58	-1.5 (-5.2 to 2.3), p=0.78	1.2 (-3.4 to 5.9), p=0.64
	Sweden	-6.1 (-26.3 to 14.2), p=0.50	-4.1 (-26.4 to 18.1), p=0.55	2.0 (-5.5 to 9.8), p=0.61
	Norway	-5.0 (-18.5 to 8.4), p=0.32	-4.4 (-17.3 to 8.6), p=0.53	0.7 (-5.0 to 5.9), p=0.81
	Nordic mean	-4.2 (-8.7 to 0.4), p=0.37	-3.2 (-7.5 to 1.0), p=0.53	0.9 (-3.6 to 5.4), p=0.68
Thaw days, Tmax $>$ 0 °C (days)	Finland	2.4 (-4.3 to 9.1), p=0.74	2.2 (-3.4 to 7.9), p=0.75	-0.2 (-6.7 to 6.3), p=0.96
	Sweden	3.9 (-5.4 to 13.2), p=0.62	4.6 (-4.3 to 13.6), p=0.50	0.8 (-6.0 to 7.7), p=0.83
	Norway	4.0 (-5.3 to 13.3), p=0.55	4.7 (-5.0 to 14.5), p=0.37	0.8 (-4.5 to 6.4), p=0.78
	Denmark	9.5 (2.4 to 16.6), p=0.36	11.1 (5.4 to 16.7), p=0.20	1.6 (-6.3 to 9.5), p=0.70
	Iceland	-2.4 (-9.0 to 4.2), p=0.68	-3.6 (-9.7 to 2.5), p=0.52	-1.2 (-6.7 to 4.3), p=0.69
	Nordic mean	2.6 (-3.3 to 8.4), p=0.67	3.1 (-2.0 to 8.2), p=0.55	0.5 (-4.7 to 5.8), p=0.83
Frost days, Tmin $<$ 0 °C (days)	Finland	-0.1 (-2.3 to 2.1), p=0.98	0.2 (-1.7 to 2.1), p=0.93	0.3 (-2.4 to 2.8), p=0.83
	Sweden	-1.2 (-9.2 to 6.9), p=0.79	-1.4 (-9.5 to 6.8), p=0.79	-0.2 (-4.8 to 4.2), p=0.94
	Norway	-2.1 (-12.0 to 7.8), p=0.68	-1.4 (-11.5 to 8.7), p=0.80	0.7 (-4.1 to 5.7), p=0.79
	Denmark	-8.6 (-24.3 to 7.0), p=0.46	-10.5 (-26.1 to 5.0), p=0.37	-1.9 (-12.6 to 8.6), p=0.73
	Iceland	6.5 (-1.4 to 14.3), p=0.43	7.2 (0.3 to 14.0), p=0.28	0.7 (-6.5 to 7.4), p=0.85
	Nordic mean	-1.1 (-6.2 to 4.0), p=0.80	-1.1 (-6.2 to 3.9), p=0.80	-0.0 (-4.0 to 4.1), p=0.99

5 Discussion

The starting hypothesis was that super El Niño winters are warmer and wetter in Finland than ordinary El Niño winters. The data give a qualified and asymmetric answer. On temperature, the three super winters were about one degree milder than the run of neutral winters of their era in Finland, Sweden, Norway and Denmark in December to February, and about 0.7 °C milder than ordinary El Niño winters, with fewer frost days, more thaw days and a little less snow. That is the direction the hypothesis predicted, but the effect is within the year-to-year noise of Nordic winters (standard deviation of detrended DJF anomalies around 2 °C in Finland), it rests on three cases, and it vanishes in January to March. On precipitation, only Finland supports the hypothesis, with a surplus of roughly a quarter of the seasonal total in all three super winters; Sweden and Denmark show none and Iceland the opposite. The absence of any relation between the continuous ONI and either metric means the super group is not the strong end of a dose-response curve.

Three features of the result deserve emphasis. First, the ordinary El Niño group is indistinguishable from neutral winters on every metric and in both windows. If the Nordic signal of El Niño is a late-winter cooling through the stratosphere, as the reviews summarise [1, 2, 4], it is too weak to see in 20 winters of national station data; the JFM Finnish value has the predicted sign and nothing more. Second, the super winters differ from ordinary El Niño winters mainly in December to February rather than in late winter, which fits the argument that the strongest events act through a different, more direct route than the moderate ones [5, 6] and cautions against extrapolating from moderate El Niño composites to the strong events. Third, the three super winters were not mild in absolute terms: Finnish DJF means of -6 to -8 °C are ordinary winters, and the 1997/98 winter was slightly colder than the 1991 to 2020 mean. Any statement that a super El Niño brings a mild Finnish winter is a statement about a residual after detrending, and the trend is the larger term.

For practical use, the seasonal forecast literature already treats the tropical Pacific as one predictor among several for European winters [7]; nothing here suggests that a super El Niño alone should change expectations for a Finnish winter by more than about one degree in DJF, with a wide margin, and by nothing detectable after New Year. The 2023/24 event, which peaked at 1.99 °C, sits at the threshold; the companion app shows that lowering the threshold to 1.8 adds it and 1972/73 to the super group and roughly halves the temperature difference. The honest summary is a modest, uncertain early-winter mildness and a Finnish precipitation surplus in three winters, both worth re-examining after the next event rather than acting on now.

6 Limitations

- Three super El Niño winters. Every statement about them is a statement about three winters, each also shaped by the NAO, the stratospheric vortex, sea ice and internal variability of that particular year. The confidence intervals show this honestly; the point estimates should not be quoted without them.
- Threshold sensitivity. Moving the super threshold from 2.0 to 1.8 adds 2023/24 and 1972/73; at 1.5 the group becomes six to eight winters. The app exposes this; the report fixes 2.0 as pre-specified.
- Detrending assumes a linear trend over each station's record. It removes part of any genuine low-frequency ENSO signal along with the warming, and it cannot remove the effect of the Atlantic Multidecadal Variability or of the 1989 to 1995 positive-NAO cluster.
- Station changes. FMI airport stations lost manual precipitation and snow observations around 2008; GHCN-Daily series contain relocations and instrument changes that national

homogenised series correct and this analysis does not. The seasonal, anomaly-based design reduces but does not eliminate the effect.

- The originally planned ERA5 coverage was reduced by API quota. Where ERA5 cross-check points exist, they are shown in the app as separate units and were not mixed with station data.
- No multiple-comparison correction; roughly 60 units, 12 metrics, 3 windows, 3 versions and 5 contrasts were tested. Isolated asterisks are expected.

7 Reproducibility

- `src/enso.py`: download ONI, define events, label winters.
- `src/fetch_fmi.py`, `src/fetch_ghcn.py`, `src/fetch_openmeteo.py`: cached raw data.
- `src/metrics.py`, `src/analysis.py`: seasonal metrics, anomalies, composites, tests.
- `src/figures.py`: figures (PDF and PNG) and the generated tables and macros used in this document.
- `src/build_app_data.py`: data bundle for `app/index.html`.

Table 4: Stations used in the national series.

Country	Station	Source	Coordinates	First	Last
Finland	Utsjoki Kevo	FMI	69.76°N, 27.01°E	1962	2026
Finland	Muonio Alamuonio	GHCN	67.97°N, 23.68°E	1959	2026
Finland	Sodankylä (GHCN)	GHCN	67.37°N, 26.63°E	1908	2026
Finland	Sodankylä Tähtelä	FMI	67.37°N, 26.63°E	1908	2026
Finland	Rovaniemi Apukka	FMI	66.58°N, 26.01°E	1959	2026
Finland	Oulu lentoasema	FMI	64.94°N, 25.34°E	1959	2026
Finland	Kajaani lentoasema	FMI	64.28°N, 27.67°E	1959	2026
Finland	Kuopio Maaninka	FMI	63.14°N, 27.31°E	1959	2026
Finland	Vaasa lentoasema	FMI	63.06°N, 21.75°E	1959	2026
Finland	Ähtäri Inha	FMI	62.55°N, 24.14°E	1959	2026
Finland	Tampere-Pirkkala	FMI	61.42°N, 23.62°E	1980	2026
Finland	Lappeenranta lentoasema	FMI	61.04°N, 28.13°E	1959	2026
Finland	Jokioinen Ilmala	FMI	60.81°N, 23.50°E	1959	2026
Finland	Turku lentoasema	FMI	60.52°N, 22.28°E	1959	2026
Finland	Helsinki Kaisaniemi	FMI	60.18°N, 24.94°E	1900	2026
Sweden	Vittangi	GHCN	67.68°N, 21.63°E	1945	2026
Sweden	Piteå	GHCN	65.32°N, 21.49°E	1859	2026
Sweden	Härnösand	GHCN	62.63°N, 17.93°E	1858	2026
Sweden	Falun	GHCN	60.62°N, 15.62°E	1860	2026
Sweden	Stockholm	GHCN	59.35°N, 18.05°E	1859	2026
Sweden	Göteborg	GHCN	57.72°N, 11.99°E	1961	2026
Sweden	Kristianstad	GHCN	56.33°N, 14.15°E	1878	2026
Norway	Tromsø	GHCN	69.65°N, 18.93°E	1920	2026
Norway	Bardufoss	GHCN	69.06°N, 18.54°E	1946	2026
Norway	Trondheim Værnes	GHCN	63.50°N, 10.89°E	1940	2026
Norway	Bergen Flesland	GHCN	60.29°N, 5.23°E	1955	2026
Norway	Oslo Blindern	GHCN	59.94°N, 10.72°E	1937	2026
Norway	Stavanger Sola	GHCN	58.88°N, 5.64°E	1940	2026
Denmark	Aalborg	GHCN	57.09°N, 9.85°E	1940	2025
Denmark	Vestervig	GHCN	56.77°N, 8.32°E	1873	2020
Denmark	København Landbohøjskolen	GHCN	55.68°N, 12.53°E	1873	2020
Denmark	Hammer Odde	GHCN	55.30°N, 14.78°E	1960	2025
Iceland	Akureyri	GHCN	65.68°N, -18.08°E	1944	2025
Iceland	Dalatangi	GHCN	65.27°N, -13.58°E	1949	2026
Iceland	Stykkishólmur	GHCN	65.07°N, -22.73°E	1949	2026
Iceland	Reykjavík	GHCN	64.13°N, -21.90°E	1941	2026

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